Homework: TypeScript Functions

100 points + 5 bonus | Due before class on September 5, 2026

Student name

Date

Part 1 - Match the Vocabulary — 15 points

Match these terms: Function, Definition, Call, Parameter, Argument, and Return.

A. A value sent into one function call

B. A reusable named job

C. A value sent back to the calling code

D. Code that describes the function and its body

E. A command that runs a function

F. A named input in the function definition

Write the six matching letters in term order:

Explain parameter versus argument in your own words:

No computer required

Predict and explain first. Use VS Code later only to check your thinking.

Predict the Output

Part 2 | 20 points

Part 2A - Count the Calls — 8 points

```
function beep() {
  console.log("Beep!");
}
beep();
beep();
```

Write the output, count the calls, and predict the no-call version:

Part 2B - Trace the Parameter — 12 points

```
function showDouble(number: number) {
  console.log(number * 2);
}
showDouble(4);
showDouble(7);
showDouble(10);
```

Write the output and the parameter value during each call:

Parameters and Arguments

Part 3 | 20 points

Part 3 - Match Inputs by Position 20 points

```
function describePet(name: string, age: number) {
  console.log(`${name} is ${age} years old.`);
}
describePet("Pixel", 3);
```

1-3. Name both parameters, their types, and their matching arguments:

4. Write two new valid calls and predict one output:

5-6. Explain both broken calls:

```
describePet(3, "Pixel");
describePet("Pixel");
```

Finish the Functions

Part 4 | 20 points

Part 4A - Complete a Greeting 6 points

```
function greetStudent(____: string) {
  console.log(`Hello, ${____}!`);
}
greetStudent("Ari");
```

Part 4B - Complete a Decision 6 points

```
function checkTemperature(temperature: ____) {
  if (temperature >= 80) {
    console.log("Hot");
  } ____ {
    console.log("Mild");
  }
}
checkTemperature(____); // choose Mild
```

Part 4C - Design Two Inputs 8 points

Write showProduct(item: string, price: number). Print one sentence and call it twice with valid arguments.

Find and Repair the Bugs

Part 5A | 15 points total

Bug A - Missing Argument 5 points

```
function welcome(name: string) {
  console.log(`Welcome, ${name}!`);
}
welcome();
```

Explain, repair, and predict:

Bug B - Wrong Argument Type 5 points

```
function showLevel(level: number) {
  console.log(`Level ${level}`);
}
showLevel("five");
```

Explain, repair, and predict:

Repair One More Bug

Part 5B | 5 points

Bug C - Defined but Never Called — 5 points

```
function announceWinner(winner: string) {
  console.log(`${winner} wins!`);
}
```

Explain why there is no output:

Add a valid call and predict the repaired output:

Debugging checklist

Find the call. Count the arguments. Match each argument type and position to its parameter.

AI Review Reflection

Part 6 | 10 points

Part 6 - One Reusable Review Rule

10 points

```
function reviewLabel(label: string) {
  if (label === "unknown") {
    console.log("Human review needed.");
  } else {
    console.log(`${label}: accepted for now.`);
  }
}
reviewLabel("tree");
reviewLabel("unknown");
```

Predict both outputs; identify the parameter and both arguments:

Why is accepted for now more accurate than guaranteed correct?

What should a person inspect when the helper flags unknown?

Core homework total: 100 points

Bonus: Return a Boolean

+5 bonus points

Bonus - Complete isPassing +5 points

```
function isPassing(score: number): boolean {
  if (score >= 70) {
    return ____;
  } else {
    return ____;
  }
}
```

Fill both blanks and predict isPassing(92) and isPassing(61):

Why does this function return a value instead of only printing one?

Return reminder

The parameter type describes what goes in. The return type describes what comes back.

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